

The Undrawn Book

Applying Basel III credit-card rules to the quarterly regulatory filings of 50 small US banks — and why most of the exposure sits where nobody currently counts it.

3,682 Basel paragraphs parsed **673** rules extracted, all grounded **600** bank-quarters computed
2023 Q1 – 2025 Q4

The brief asks for four things in order: extract the rules, classify them, derive parameters from them, then find those parameters in banks' quarterly reports. This document covers that chain as it now stands, end to end, with a worked example. Everything described here runs — the figures below are produced by the code, not estimated.

01 – Inputs

Two corpora, and the join between them

The project needs a regulatory side and a bank side, and the difficulty the supervisor identified is real: the quarterly report is structured, but it is not written in the vocabulary of the rules.

Regulatory side

- BIS consolidated Basel Framework, 1,982 pages, parsed to **3,682 paragraphs**, each keeping its real citation (`CRE20.68`) and heading path.
- **673 rules** extracted across 23 chapters — 367 with a number in them, 144 bearing on cards.
- Classified by kind: capital, liquidity, credit-risk RWA, concentration, provisioning, issuance, disclosure.

Bank side

- FFIEC Call Reports — the mandatory quarterly filing — pulled from the CDR bulk distribution.
- **50 banks × 12 quarters = 600 filings**, complete panel, no gaps.
- Small banks: under \$10bn in assets, the community bank leverage-ratio threshold. Median \$4.1bn.
- Each filing is ~1,750 fields keyed by MDRM code (`RCON3815`).

The join

A rule cites a Basel paragraph; a filing reports an MDRM code. The bridge is a fixed vocabulary of named quantities, each bound to specific codes — so a formula written against a rule is executable against a filing. A formula naming `EAD_transactor` cannot be run against anything; one naming `credit_card_balances` resolves to `RCFDB538` or `RCONB538` and produces a number.

02 — Method

Five steps, each resting on a cited rule

The steps are a genuine sequence: each consumes the previous one's output. The rail on the left of each step names the provisions it rests on.

1

CRE20.94

CRE20.100

12 CFR 217.33

Exposure at default

Basel converts off-balance-sheet commitments into credit-equivalent exposure with a credit conversion factor. A commitment that is *unconditionally cancellable at any time without prior notice* takes a **10% CCF** — which is exactly what a credit-card line is.

$$\text{EAD} = \text{drawn} + \text{undrawn} \times \text{CCF}$$

Comenity: $\$6.76\text{bn} + \$37.77\text{bn} \times 0.10 = \mathbf{\$10.53\text{bn}}$

The same undrawn lines take a **0% CCF** under the US rule in force today, so they contribute nothing to reported exposure.

2

CRE20.65

CRE20.66

CRE20.68

12 CFR 217.32

Risk weight

Credit cards are named explicitly in Basel's "regulatory retail" product criterion, so the book qualifies subject to a €1m per-counterparty ceiling and a 0.2% granularity limit. The weight then turns on whether the obligor is a *transactor* — someone who repaid in full at every scheduled date for twelve months.

45% transactor · **75%** regulatory retail, non-transactor · **100%**
other retail

No US regulatory report collects the transactor split. The output is therefore a **band across all three weights**, not a point estimate — with 75% reported as central, since a bank that cannot demonstrate transactor status cannot claim 45%.

3

RBC20.1

RBC30.2

RC-R P859

RC-R A223

Capital adequacy

The card book's Basel RWA replaces its US contribution in the bank's reported total, and the restated ratio is tested against the minimum and the buffer.

$$RWA' = RWA_{reported} - (drawn \times 100\%) + (EAD \times rw)$$

test **CET1** $\geq 4.5\%$ (minimum) and $\geq 7.0\%$ (with the 2.5% conservation buffer)

Falling into the buffer range constrains capital distributions rather than breaching the minimum, so the two thresholds are reported separately.

4

LCR40.64

RC 0071 / 0081

RC 1773 , JJ34

Liquidity stress

Basel's LCR assumes a 5% **drawdown** within thirty days on undrawn committed facilities to retail and small-business customers. Set against the bank's cash, securities and money-market placements, this is the supervisor's own worked example made computable: *can the bank fund the lines it has already promised?*

$$\text{cover} = \text{liquid assets} \div (\text{undrawn} \times 0.05)$$

Comenity: $\$1.23\text{bn} \div \$1.89\text{bn} = 0.65\times$

This is an *analytical stress*, not a compliance test — the US LCR rule does not reach banks of this size. That distinction is stated wherever the number appears.

5

RC-N B575 / B576

RI-B B514 / B515

RI B485

RC-K B561

Credit quality and economics

Delinquency, loss rate, portfolio yield and allowance coverage — computed against the *average* balance rather than period-end, since income items are flows.

Comenity: 30–89 days **3.4%** · 90+ days **3.6%** · net charge-offs **7.0%** · yield **26.8%** · allowance/drawn **11.4%**

These are the inputs to the performance criterion the brief asks us to define, and they are what a redistribution proposal has to preserve.

03 – Worked example

Comenity Bank, year-end 2025

A \$7.8bn Delaware bank whose loan book is 100% credit cards — and whose committed but undrawn lines are five times its balance sheet.

Same bank, same filing, four regulatory treatments. All figures computed from the 2025-12-31 Call Report.

TREATMENT	EAD	CARD RWA	TOTAL RWA	CET1 RATIO	STATUS
US rule in force today	\$6.76bn	\$6.76bn	\$6.98bn	15.1%	above buffer
Basel @ 45% — all transactors	\$10.53bn	\$4.74bn	\$4.96bn	21.2%	above buffer
Basel @ 75% — central case	\$10.53bn	\$7.90bn	\$8.12bn	13.0%	above buffer
Basel @ 100% — other retail	\$10.53bn	\$10.53bn	\$10.76bn	9.8%	above buffer

A separate figure answers a different question: converting the undrawn lines alone, while leaving the drawn book at its current US treatment, gives **10.7%**. That isolates the exposure nobody counts today. The 13.0% row is what Basel would actually require, because Basel also moves the drawn balances from 100% down to 75%.

What the example demonstrates

The bank is comfortably capitalised on every measure — and its liquidity cover on its own committed lines is **0.65×**. The binding constraint is not capital. It is that a 5% drawdown of promises already made would exhaust the cash and securities on hand. That is the question the supervisor posed, answered with the bank's own filing.

04 – Coverage

Across all 50 banks

Selected results at 2025-12-31. Ordinary community banks barely move — their card books are small. The specialists move enormously.

BANK	ASSETS	UNDRAWN LINES	CET1 REPORTED	BASEL @ 75%	LIQUIDITY COVER
Credit First NA	\$0.04bn	\$11.44bn	387.7%	4.3%	0.06×
Comenity Bank	\$7.76bn	\$37.77bn	15.1%	13.0%	0.65×
John Deere Financial	\$3.40bn	\$33.50bn	—	CBLR	0.00×
Credit One Bank	\$1.97bn	\$8.11bn	—	CBLR	0.53×
Nicolet National Bank	\$9.17bn	\$0.14bn	11.2%	11.1%	—
Peoples Bank	\$9.64bn	\$0.09bn	12.0%	12.0%	—

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banks cannot cover a 5% drawdown of their own committed card lines from cash and securities

11.9 → 14.3%

line utilisation across the cohort, 2023 to 2025, while committed capacity stayed flat at ~\$130bn

05 – Limits

What is not computable, and why that matters

The most useful output of a data-grounded method is an honest account of where it stops. Three limits are structural, not fixable by better engineering, and each is a finding in its own right.

Availability of the quantities Basel's card rules require, across the 50 banks at 2025-12-31.

QUANTITY	SOURCE	COVERAGE	CONSEQUENCE
Unused card lines	RC-L 3815	49 / 50	usable
Card balances	RC-C B538	45 / 50	usable
Card interest income	RI B485	43 / 50	usable
Delinquency 30–89 / 90+	RC-N B575/B576	37 / 26	partial
Risk-weighted assets	RC-R A223	42 / 50	8 CBLR filers report none
Transactor / revolver split	—	0 / 50	not collected anywhere
Card book by PD band	RC-O N040–N054	0 / 50	large filers only
HQLA, net 30-day outflows	FR 2052a	0 / 50	\$100bn+ filers only

Why each limit is itself a result

The transactor split decides Basel's headline retail weight and no one collects it. A rule that turns on an unobservable is unenforceable from outside the issuer — which is why the output is a band.

Eight of fifty banks report no risk-weighted assets at all. Having elected the community bank leverage ratio, they are outside the risk-based framework entirely. Basel's risk weights are not hard to check for them; they do not apply. That is the supervisor's own point — requirements scale with size — showing up as missing data.

The data that would permit a proper risk segmentation exists, and small banks are exempt from reporting it. Schedule RC-O buckets the card book by two-year probability of default; it is blank for every bank under \$10bn. So the analysis must use the standardised approach, not the internal-ratings approach.

The divergence to settle before the next phase

Basel and the US rule disagree on precisely the provisions this project centres on. Basel splits regulatory retail 45%/75% by transactor status; the US applies a flat 100% and does not recognise the split. Basel converts unconditionally cancellable commitments at 10%; the US converts them at 0% — so the \$128.6bn of undrawn card lines across these 50 banks attracts no capital at all under the rules they actually report against.

Applying Basel literally to Call Report data will therefore not reproduce any US bank's published ratios. The question for you: extract Basel as specified, the US implementation the filings reflect, or both with an explicit divergence map? The third is more work and is the only one that makes the comparison defensible.

06 — Assurance

How we know the numbers are right

Rules are grounded mechanically. Every extracted rule carries a quotation that must appear character-for-character in the paragraph it cites, checked by exact substring match. No model is asked whether a quote is faithful, so no model can be wrong about it. All 673 rules pass; those that fail are dropped, not repaired.

Field mappings are cross-validated. The CET1 ratio computed from our own mapping reproduces the ratio each bank reports itself. Where it did not, the mapping was wrong and was fixed.

Regulatory assumptions are tested against the data, not asserted. Comenity's reported RWA divided by its drawn balances is 1.03 — RWA is essentially the drawn book at 100%, and \$37.8bn of undrawn lines contributes nothing. That confirms the US 0% conversion factor empirically rather than on the strength of a citation.

Implausible results are chased, not shipped. John Deere Financial appeared to hold zero liquid assets; the cause was a field that not every bank files, and correcting it changed the population failing the liquidity stress from seven banks to six.

Status of the US-side claims

The Basel citations above are drawn from the parsed framework and are checked. The claims about the US rule — the 0% conversion factor, the flat 100% retail weight, the \$10bn and \$100bn thresholds — are stated from the regulation but a verification pass against the Code of Federal Regulations is still running. US adoption of the Basel III endgame has been proposed and revised repeatedly, so *in force today* versus *proposed* needs confirming before these go into anything final.

What this sets up

Steps 1–5 produce a comparable, rule-grounded profile for every bank in every quarter. That is the input the rest of the brief needs.

Performance criterion and ranking. Yield, loss rate, capital headroom and liquidity cover are all computed; what remains is choosing and defending a weighting, then ranking within the size tier.

Benchmark and redistribute. The cohort is already usefully split — Comenity withdrew \$14.2bn of committed lines over three years while Merrick grew its book 93%. Those are opposed strategies at comparable scale, which is better material than a peer group that all behaves alike.

Scenarios. The five steps are the objective function a scenario perturbs: a utilisation shock moves step 1, a migration to "other retail" moves step 2, a funding shock moves step 4. Each is bounded by something observed in the panel, which is the plausibility check the brief requires.

Preliminary methodology · Basel III applied to small-bank credit-card portfolios · Sources: BIS consolidated Basel Framework; FFIEC Call Reports via the CDR public bulk distribution, 2023 Q1 – 2025 Q4. All figures computed from the filings; monetary amounts in US dollars.